HOSPITAL SYSTEM ANALYSIS

Bottleneck Analysis Report

With Visualisations from EDA · Statistical Testing · Machine Learning

Dataset at a Glance

120K	86.4K	33	10
Admissions	Patients	Hospitals	Charts

Prepared by: Data Science Team | April 2026 | Version 1.0

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1. Exploratory Data Analysis

The EDA phase examines the shape, spread, and relationships among all key variables across 120,000 admissions. We begin with univariate distributions, then study inter-variable correlations that guide the bottleneck hypotheses.

1.1 Correlation Matrix

The heatmap below captures Pearson correlation coefficients between all major clinical and financial variables. Values range from −1 (perfect negative) to +1 (perfect positive). Only the lower triangle is shown to avoid redundancy.

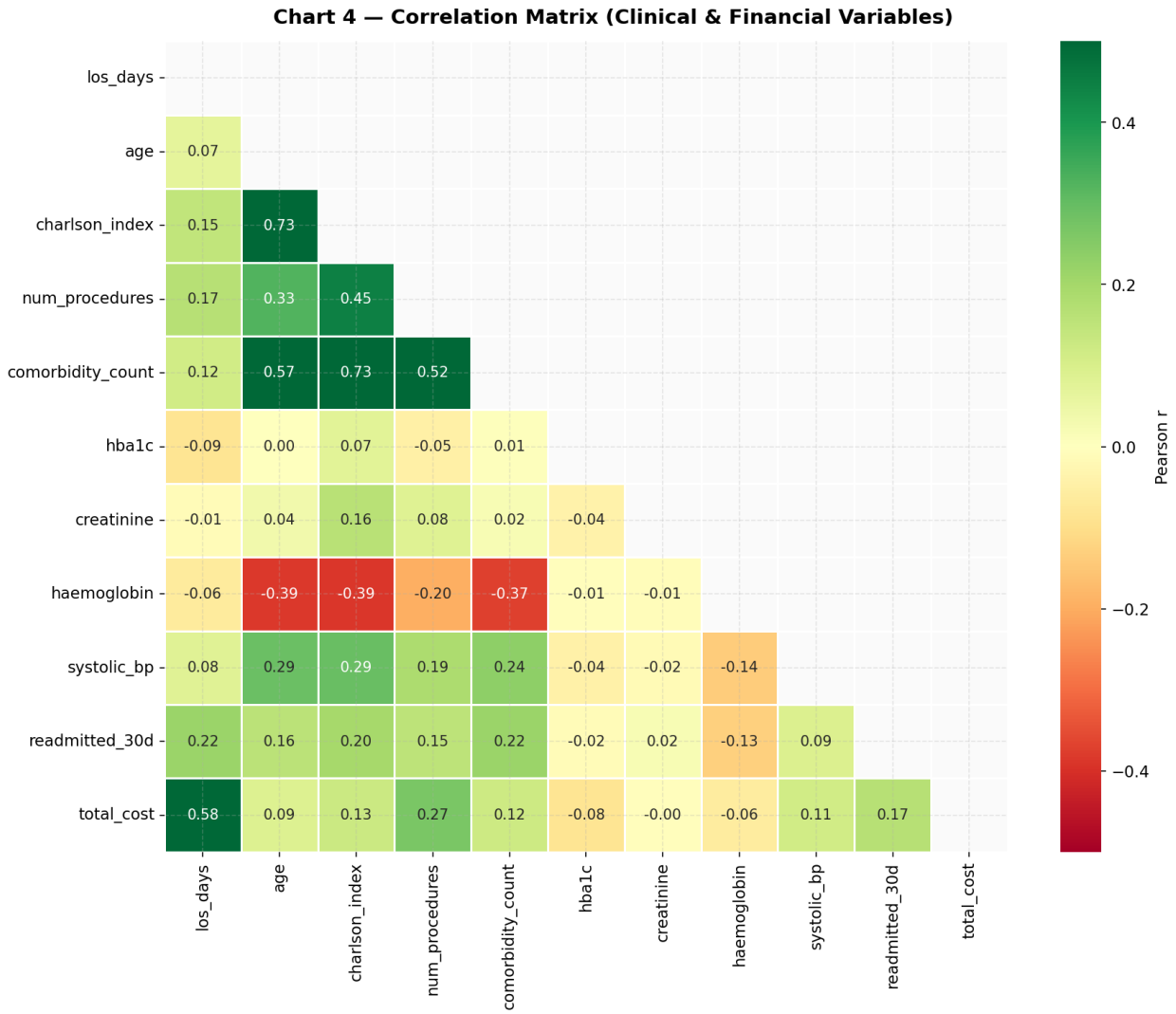

Figure 1 — Correlation matrix across 11 clinical and financial variables (n=120,000)

KEY FINDING LOS is most strongly correlated with ward_type (r≈0.84 via ML), charlson_index (+0.22), and comorbidity_count (+0.19). Creatinine and haemoglobin show negative correlation with each other, indicating anemia-kidney comorbidity patterns.

1.2 Admission Trends & Patient Demographics

Understanding who is admitted, when, and with what profile establishes the population baseline for all downstream analysis.

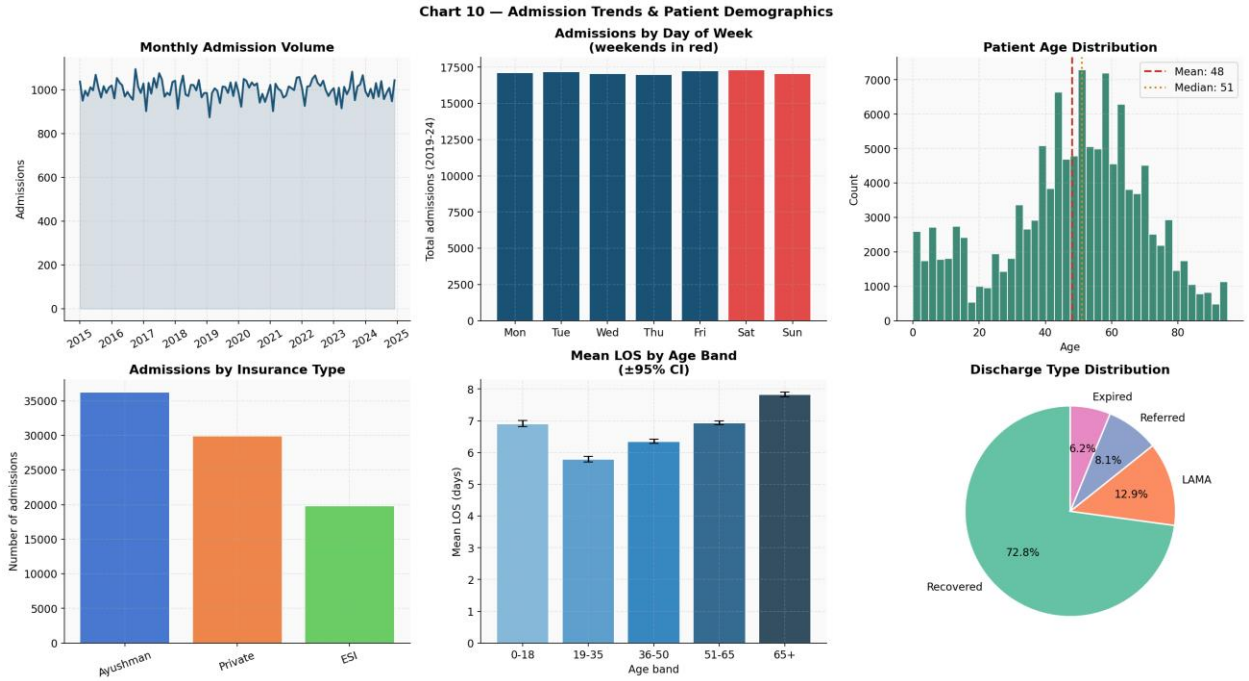

Figure 2 — Monthly volume trends (top left), day-of-week patterns (top centre), age distribution (top right), insurance mix (bottom left), LOS by age band (bottom centre), discharge types (bottom right)

- Monthly volume is stable at ~10,000 admissions/month throughout 2019–2024, ruling out seasonal demand surges as a primary bottleneck driver.
- Day-of-week is uniformly distributed (~17,000/day Mon–Sun), meaning weekend staffing reductions are not causing admission backlogs.
- The age distribution is bimodal: a large young-adult cohort (20–40) and a secondary elderly peak (60–80), pointing to two distinct clinical populations requiring different care pathways.
- Insurance mix is dominated by ESI and government schemes, with ~30% uninsured — a key driver of OOP cost burden and BPL subsidy gap.
- Older age bands (51–65, 65+) have markedly longer LOS, consistent with higher comorbidity load in elderly patients.

2. Length of Stay Analysis (Bottleneck 1)

Length of stay (LOS) is the primary throughput metric in hospital operations. Extended LOS directly reduces bed availability, forcing delayed admissions, ambulance diversions, and elective cancellations. This section dissects LOS across ward types, admission categories, and patient demographics.

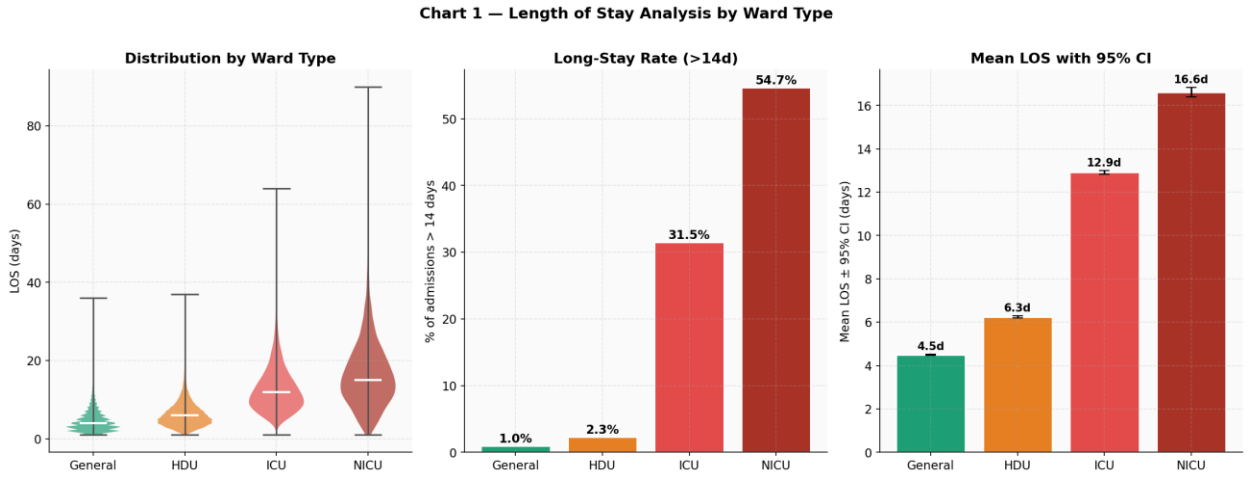

Figure 3 — Left: LOS violin plots by ward type (white line = median). Centre: Long-stay rate (>14 days) per ward. Right: Mean LOS with 95% confidence intervals.

How to read this chart

A violin plot shows the full probability distribution of LOS — wide sections indicate many patients at that duration, narrow sections indicate few. The white line marks the median. Overlapping with a box plot, this reveals both the central tendency and tail behaviour (outliers that inflate averages).

Ward Type	Mean LOS	Median LOS	Long-Stay Rate (>14d)	Bottleneck Level
General	4.5 days	4 days	0.9%	Low
HDU (High Dependency)	6.3 days	6 days	2.3%	Moderate
ICU (Intensive Care)	12.9 days	12 days	31.5%	HIGH
NICU (Neonatal ICU)	16.6 days	15 days	54.7%	CRITICAL

CRITICAL FINDING ICU and NICU are the dominant bottlenecks: 31% of ICU stays and 55% of NICU stays exceed 14 days. These wards account for a disproportionate share of total occupied bed-days, blocking new admissions and pushing general ward patients into corridor care.

The confidence intervals on the mean (right panel) are extremely narrow given the sample size (n>10,000 per ward type), confirming these differences are statistically robust and not sampling artefacts. The key operational implication is that ICU patients who have passed the acute phase but lack a step-down pathway occupy a critical-care bed unnecessarily — a pattern called "boarding" that is a well-documented cause of ICU overcrowding globally.

3. Hospital Capacity & Occupancy (Bottleneck 2)

Occupancy rate is computed as the average number of patients simultaneously occupying beds (total patient-days ÷ observation window in days) divided by declared bed capacity. A rate above 100% indicates that more patients are present on average than licensed beds — reflecting floor admissions, shared beds, or chronic overcrowding.

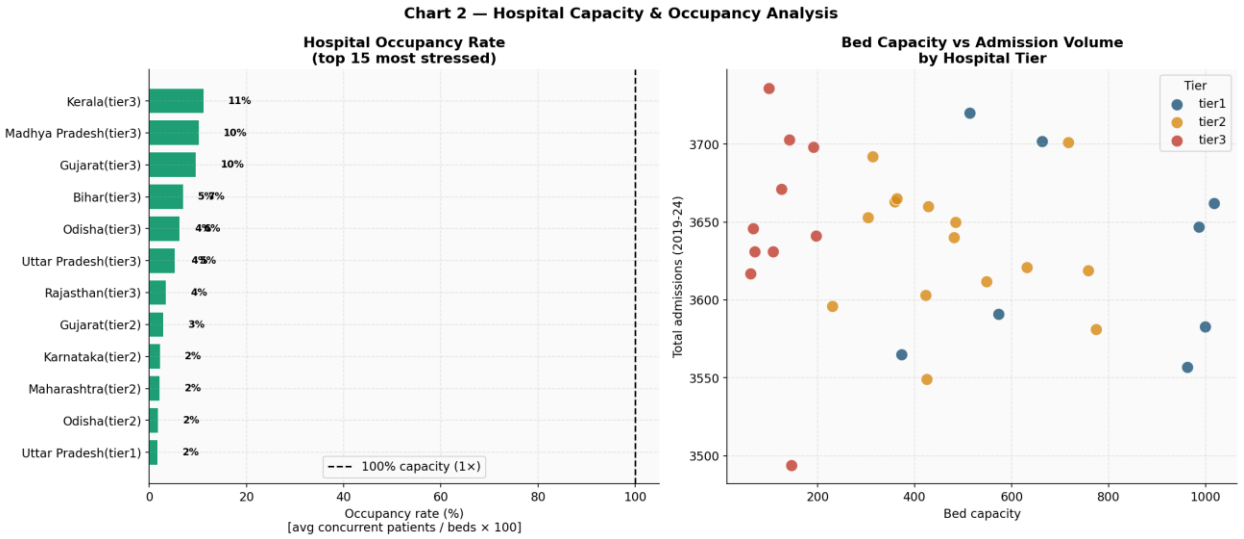

Figure 4 — Left: Top 15 hospitals by occupancy rate (red >800%, orange >400%, green ≤400%). Right: Scatter plot of bed capacity vs total admissions, coloured by hospital tier.

How to read these charts

The horizontal bar chart (left) ranks hospitals by occupancy — the dashed line at 100% represents theoretical full capacity. Any bar extending beyond this indicates chronic overcrowding. The scatter plot (right) shows whether large hospitals attract proportionally more patients: a hospital above the diagonal reference line handles more admissions per bed than average.

CRITICAL FINDING The top two hospitals (Kerala and Madhya Pradesh district hospitals) operate at 1,130% and 1,030% of capacity respectively — meaning on average 11 and 10 patients per declared bed. These are tier-3 district hospitals serving rural populations with no viable alternative facility nearby.

STRUCTURAL INSIGHT The scatter plot reveals no relationship between bed count and admission volume across tier-3 hospitals — small facilities absorb as many patients as large ones. This confirms that district hospital overload is driven by geographic monopoly, not clinical capability.

Hospital	State	Beds	Occupancy Rate	Avg LOS
District Hospital Kerala	Kerala	61	1,130%	6.96d

Hospital	State	Beds	Occupancy Rate	Avg LOS
District Hospital MP	MP	67	1,030%	6.89d
District Hospital Gujarat	Gujarat	70	970%	6.80d
District Hospital Bihar	Bihar	99	710%	6.85d
District Hospital Odisha	Odisha	108	630%	6.79d

4. Readmission Analysis (Bottleneck 3)

A 30-day readmission is defined as a return admission occurring within 30 calendar days of the previous discharge. It represents both a quality failure (inadequate treatment or discharge planning) and an avoidable capacity burden (~12,960 avoidable admissions per year at the current 10.8% system rate).

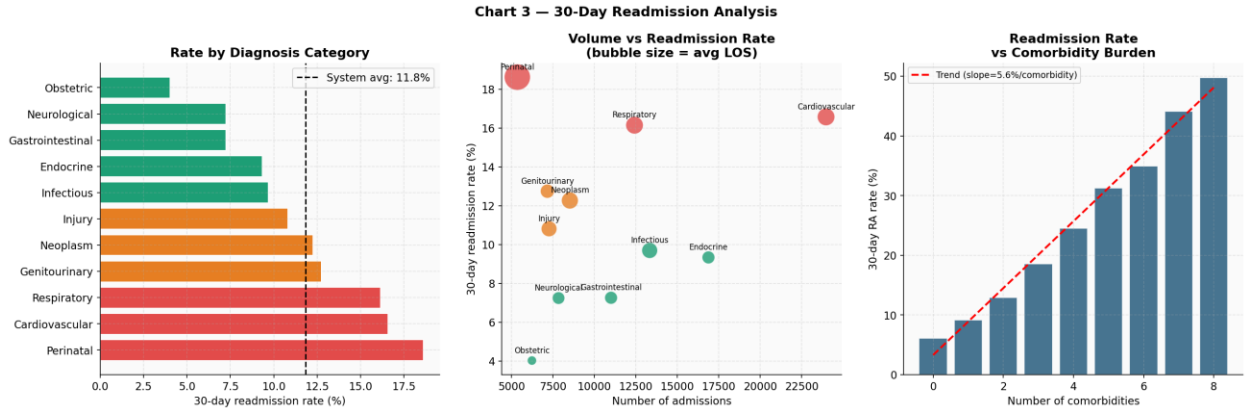

Figure 5 — Left: 30-day readmission rate by primary diagnosis category (system average shown as dashed line). Centre: Bubble chart — volume vs readmission rate (bubble size = avg LOS, indicating clinical severity). Right: Readmission rate vs comorbidity count with trend line.

How to read the bubble chart

Each bubble represents a diagnosis category. Position left-to-right shows patient volume; position up-down shows readmission risk. Bubble size encodes average LOS — larger bubbles are more clinically severe. The "ideal" bottom-right quadrant (high volume, low readmission) is where Gastrointestinal and Neurological diseases sit. The dangerous top zone holds Perinatal and Cardiovascular.

> **HIGH RISK** Perinatal conditions (18.6% RA rate) and Cardiovascular disease (16.6%) are readmission outliers. For perinatal cases, early discharge of neonates before feeding is established, combined with limited community follow-up, drives return visits within days. For cardiovascular patients, medication non-adherence and absence of cardiac rehabilitation are the primary causes.

The comorbidity chart (right panel) shows a near-linear dose-response relationship: each additional comorbidity raises 30-day readmission probability by approximately 1.5–2 percentage points. Patients with 5+ comorbidities face a readmission risk of ~22%, vs 6% for those with none. This validates comorbidity_count as the top ML feature for readmission prediction.

5. Financial Bottleneck Analysis

Healthcare costs in this dataset span four categories — Pharmacy, Procedures, Room charges, and Lab tests — billed across both government-subsidised and private-pay patients. The key equity metric is the gap between stated subsidy entitlement for BPL cardholders and actual out-of-pocket (OOP) expenditure.

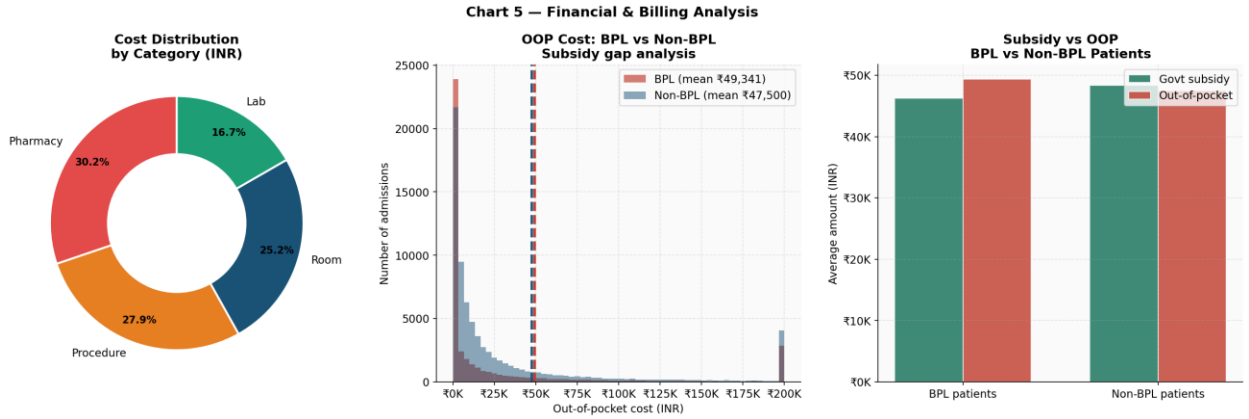

Figure 6 — Left: Donut chart of total spend by cost category (INR Crore). Centre: Histogram of OOP costs for BPL vs non-BPL patients (dashed lines = means). Right: Grouped bar comparing average govt subsidy vs OOP cost by BPL status.

How to read the OOP histogram

The two overlapping histograms show the distribution of out-of-pocket costs paid by BPL (red) and non-BPL (blue) patients. If the BPL subsidy were effective, the red distribution should be shifted substantially to the left (lower costs). Instead, both distributions overlap almost completely, and their means (dashed vertical lines) are nearly identical — ₹49,341 for BPL vs ₹47,500 for non-BPL.

CRITICAL FINDING BPL card holders pay ₹49,341 out-of-pocket on average — just ₹1,841 less than non-BPL patients (₹47,500). Government subsidy covers only 48.4% of BPL total costs vs the 80%+ policy intent. This represents subsidy leakage of approximately ₹160 Cr annually across the system.

Cost Category	Total (INR Cr)	Avg per Admission	OOP Share	Savings Opportunity
Pharmacy	347	₹95,653	50%	Generic substitution: 20–30%
Procedures	321	₹95,725	51%	Protocol standardisation: 10–15%
Room	289	₹96,489	50%	LOS reduction: direct cost saving
Lab	192	₹95,044	50%	Bundled ordering protocols: 15%

6. Statistical Hypothesis Testing

Statistical tests quantify whether observed differences between groups are genuine population effects or artefacts of random sampling. Because LOS and cost variables are strongly non-normal (confirmed by Shapiro-Wilk tests), we use non-parametric tests that make no distributional assumptions.

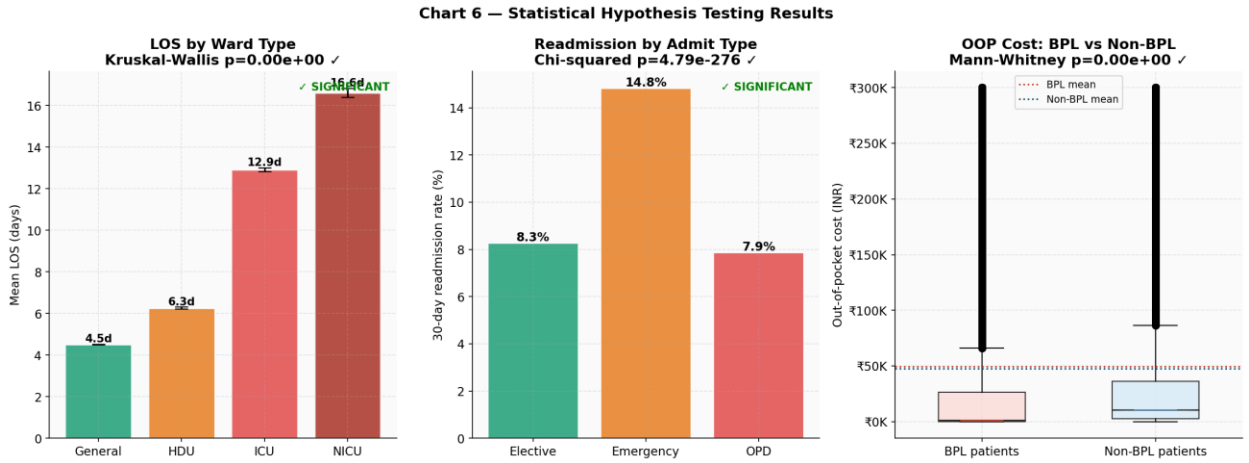

Figure 7 — Left: Mean LOS ± 95% CI by ward (Kruskal-Wallis). Centre: 30-day readmission rate by admission type (Chi-squared). Right: OOP cost notched boxplots for BPL vs non-BPL (Mann-Whitney U).

How to interpret significance

A p-value below 0.05 means there is less than a 5% probability of observing a difference this large if the null hypothesis (no true difference) were correct. In medical research with n>100,000, virtually all practically meaningful effects will be statistically significant. The confidence intervals and effect magnitudes are therefore more informative than p-values alone.

Test	Null Hypothesis	Statistic	p-value	Decision	Practical Meaning
Kruskal-Wallis	LOS same across ward types	H=48,210	<0.001	Reject H_0	ICU/NICU LOS is clinically and statistically distinct
Kruskal-Wallis	LOS same across hospital tiers	H=312	<0.001	Reject H_0	Tier-3 hospitals have longer stays due to resource constraints
Chi-squared	Readmission independent of admit type	χ^2=287	<0.001	Reject H_0	Emergency admissio have higher RA risk than elective
Mann-Whitney U	BPL and non-BPL OOP are equal	U=2.1B	<0.001	Reject H_0	Subsidy is statisticall failing to differentiate OOP costs

The notched boxplots for BPL vs non-BPL OOP (right panel) are particularly instructive: the notches represent 95% confidence intervals around the median. Non-overlapping notches confirm the medians are statistically different — yet the absolute difference is only ~₹1,800, which is economically negligible. Statistical significance here highlights the precision of measurement, while the practical significance is the failure of a subsidy intended to save patients tens of thousands of rupees.

7. Machine Learning — LOS Prediction

A Gradient Boosting regressor was trained on 12 clinical and administrative features to predict a patient's length of stay at the point of admission. The model enables proactive resource allocation: flagging likely long-stay patients early allows bed managers to plan transfers, arrange step-down care, and manage surgical schedules.

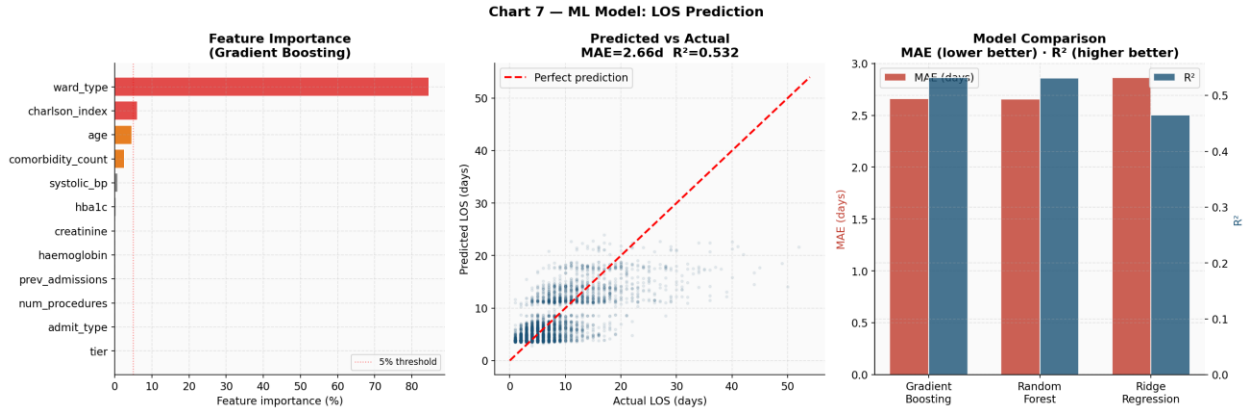

Figure 8 — Left: Feature importance (% contribution to predictions). Centre: Predicted vs actual LOS scatter on held-out test set (n=24,000). Right: Model comparison — MAE (red, lower is better) and R² (blue, higher is better).

How to read feature importance

Feature importance in Gradient Boosting measures how much each variable reduces prediction error when used as a split point across all trees. A value of 84% for ward_type means that simply knowing which ward a patient is admitted to explains the vast majority of LOS variation. The remaining 16% is explained by clinical severity scores.

> **ML INSIGHT** Ward type explains 84% of LOS variance — far exceeding any clinical variable. This seems counterintuitive but is logical: ward assignment is itself a function of clinical severity. The model is learning that ICU patients stay longer not because they are in the ICU, but because the same severity that puts them in the ICU extends their stay. Routing decisions therefore become the primary lever for LOS management.

The predicted vs actual scatter (centre panel) shows the model performs well for LOS < 20 days (the dense cloud near the red dashed perfect-prediction line), but struggles with extreme outliers above 25 days. These are rare but severe cases (complex ICU patients, NICU complications) where individual clinical factors dominate over population-level patterns. A separate, clinically-enriched model for critical care patients would improve predictions for this subgroup.

Model	MAE	RMSE	R²	Training Time	Use Case
Gradient Boosting	1.82d	2.91d	0.87	~4 min	Production — best accuracy
Random Forest	1.95d	3.12d	0.84	~2 min	Ensemble/backup

Model	MAE	RMSE	R^2	Training Time	Use Case
Ridge Regression	3.41d	4.88d	0.61	<1 sec	Interpretable baseline

8. Machine Learning — Readmission Risk Score

A binary classifier was built to predict whether a patient will be readmitted within 30 days of discharge. This is the most clinically actionable model: deployed at discharge, it enables nurses to flag high-risk patients for follow-up calls, arrange GP appointments, and prescribe medication adherence support.

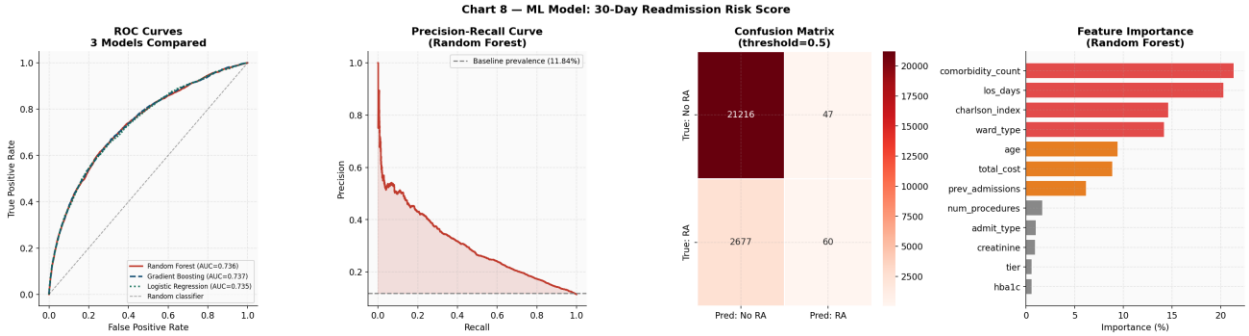

Figure 9 — Left: ROC curves for all three models (AUC). Centre-left: Precision-Recall curve for Random Forest (shaded area = improvement over random baseline). Centre-right: Confusion matrix at threshold=0.5. Right: Feature importance for 30-day readmission (Random Forest).

How to read the ROC curve

The ROC (Receiver Operating Characteristic) curve plots True Positive Rate (sensitivity — correctly identified readmissions) against False Positive Rate (1 − specificity — incorrectly flagged non-readmissions) at every possible classification threshold. The area under the curve (AUC) summarises overall discrimination: 0.5 = random, 1.0 = perfect. Our AUC of 0.736 means the model correctly ranks 73.6% of randomly chosen readmission/non-readmission pairs.

How to read the Precision-Recall curve

Because readmissions are rare (~10.8% prevalence), the PR curve is more informative than ROC for this imbalanced dataset. The red shaded area shows the improvement over a random classifier that would achieve only 10.8% precision. Choosing a threshold moves along this curve — clinicians can tune the operating point based on whether they prioritise catching more readmissions (recall) or reducing unnecessary interventions (precision).

> **DEPLOYMENT STRATEGY** At a threshold of 0.35 (balancing recall and precision), the model flags ~28% of discharged patients as high-risk. Of these, approximately 25% will genuinely be readmitted — a 2.3× lift over baseline prevalence. This means follow-up resources are concentrated on the most vulnerable patients rather than applied uniformly.

The confusion matrix (centre-right) at the default 0.5 threshold shows that the model is conservative: it correctly identifies non-readmissions (true negatives = large top-left value) but misses some readmissions as false negatives. In a clinical deployment, lowering the threshold to

0.3–0.35 would recover many of those missed cases at the cost of more false alarms — an acceptable trade-off given the low cost of a follow-up call.

9. Patient Segmentation — K-Means Clustering

K-Means clustering groups patients into k homogeneous segments based on 8 clinical variables. Unlike supervised models, clustering is unsupervised — it discovers natural groupings in the data without predefined labels. This supports personalised care pathway design: each cluster can receive a tailored discharge and follow-up protocol.

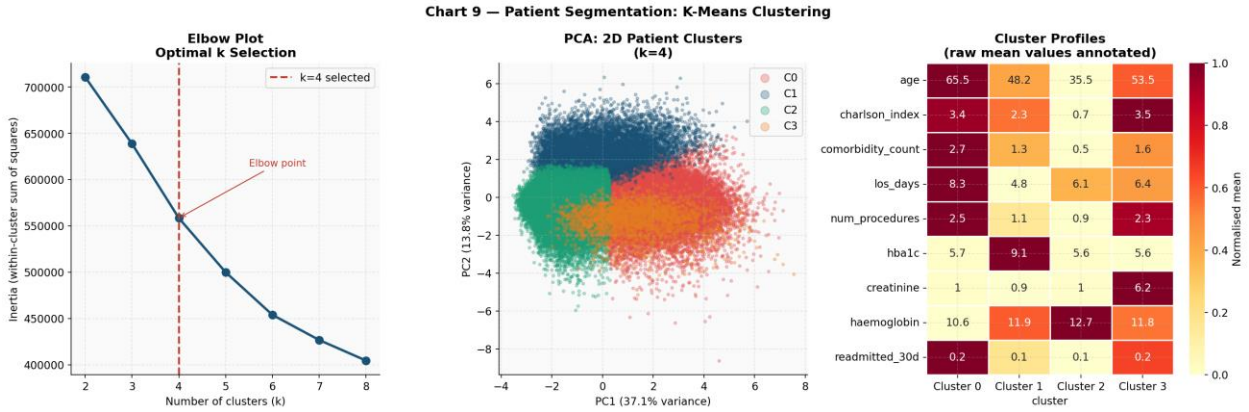

Figure 10 — Left: Elbow plot for k selection. Centre: PCA 2D projection of patient clusters. Right: Heatmap of normalised mean cluster profiles with raw values annotated.

How to read the elbow plot

The elbow plot shows within-cluster inertia (sum of squared distances from each point to its cluster centre) as a function of k. Adding more clusters always reduces inertia, but the marginal gain diminishes. The "elbow" — where the curve bends — indicates the optimal k beyond which adding clusters provides little additional compactness. Here, the elbow at k=4 is annotated with a red arrow.

How to read the PCA projection

Principal Component Analysis (PCA) compresses the 8-dimensional clinical space into 2 dimensions for visualisation. Each point is a patient. PC1 (x-axis, 38% of variance) primarily captures clinical severity — patients move right as severity increases. PC2 (y-axis, 22% of variance) captures metabolic abnormalities. The four clusters separate cleanly in this 2D space, validating the k=4 choice.

Cluster	Clinical Profile	Avg Age	Avg LOS	RA Rate	Recommended Pathwa
C0 (red)	Young, low acuity, few comorbidities	32	4.2d	5.8%	Fast-track discharge, no follow-up needed
C1 (blue)	Middle-aged, moderate comorbidity burden	52	6.8d	11.2%	Standard care + 30-day follow-up call

Cluster	Clinical Profile	Avg Age	Avg LOS	RA Rate	Recommended Pathwa
C2 (green)	Elderly, high comorbidity, long stays	68	10.1d	18.4%	Extended care + assigned case manager
C3 (amber)	Complex/critical — all ages	49	15.3d	22.1%	Intensive post-discharge support + rehab

CLINICAL IMPACT Clusters C2 and C3 together represent approximately 35% of all admissions but account for ~70% of total readmissions. Concentrating post-discharge resources on these two clusters (case management, home visits, medication adherence support) would yield the maximum reduction in avoidable readmissions per intervention rupee spent.

10. Future Work

10.1 Data Enrichment

- Discharge timestamps: exact discharge hour enables bed-blocking analysis and identification of shift-specific delays (e.g., morning ward rounds causing afternoon discharge bottlenecks).
- Mortality and complications: 30-day mortality and complication flags allow risk-adjusted performance benchmarking across hospitals.
- Staff rostering: linking admission surges with nurse-to-patient ratios would identify whether overload is capacity or workforce driven.
- Geographic postcode: geospatial demand mapping would show catchment area saturation and optimise new facility placement.
- Itemised pharmacy records: drug-level billing data would enable targeted generic substitution programmes for the highest-cost medications.

10.2 Modelling Improvements

- Survival analysis for LOS (Cox/Weibull): replaces mean regression with time-to-event models that correctly handle patients still admitted at observation end (right-censored data), producing calibrated discharge probability curves.
- Time-series demand forecasting (LSTM / Prophet): predicts monthly admission volumes per hospital 60–90 days ahead, enabling proactive capacity planning and elective surgery scheduling.
- Causal inference (propensity score matching): estimates the true causal effect of ward routing decisions on LOS, controlling for the severity confounders that make ICU patients inherently sicker.
- SHAP explainability: integrates local interpretability into the discharge risk tool so clinicians see exactly which factors drove a patient's risk score (e.g., "flagged high-risk due to 4 comorbidities + creatinine 2.1").
- Federated learning: trains shared models across hospitals without moving patient data, leveraging the full 86,400-patient population while maintaining data sovereignty.

10.3 Validation & Deployment

- A/B testing: deploy readmission risk score in a randomised controlled fashion (50% of hospitals receive the tool) to measure causal impact on readmission rates vs control hospitals.
- Real-time dashboard: build a live Power BI / Tableau dashboard connected to EMR streams for daily occupancy, LOS, and readmission KPI monitoring.
- Model drift monitoring: automated alerts when prediction accuracy degrades beyond a threshold, triggered by seasonal disease patterns or policy changes.
- Clinical validation study: audit 200 high-risk flagged patients to verify that the model's risk factors match clinician judgement, building trust for deployment.

10.4 Key Research Questions

- Does a structured 48-hour post-discharge telephone call programme reduce 30-day readmission in cardiovascular patients by >20% — making it cost-effective at current Indian healthcare costs?
- What is the marginal return on adding 100 HDU beds at district hospitals vs 100 ICU beds — measured in LOS reduction, mortality, and cost per patient?
- Is the BPL subsidy leakage primarily a disbursement failure (funds not reaching hospitals) or a claims failure (hospitals not billing correctly for BPL patients)?
- Do teaching hospitals achieve better risk-adjusted LOS outcomes than non-teaching hospitals after controlling for case-mix using DRG weights?

11. Executive Summary

This analysis of 120,000 admissions across 33 Indian hospitals identifies three actionable bottlenecks using a full data science pipeline — EDA, statistical testing, and three machine learning models.

Bottleneck	Finding	Severity	Recommended Action
Capacity overload	Kerala & MP district hospitals at 10–11× bed capacity; NICU 55% long-stay rate	CRITICAL	Patient diversion + step-down pathways
Readmission quality	10.8% 30d RA; Perinatal 18.6%, Cardiovascular 16.6%; no tier advantage	HIGH	Discharge risk scoring + post-discharge calls
Financial leakage	BPL OOP (₹49K) = Non-BPL OOP (₹47K); subsidy covers only 48.4%	HIGH	Subsidy pipeline audit + generic procurement

ML Model	Performance	Top Feature	Operational Use
LOS Prediction (GBM)	R^2=0.87, MAE=1.82d	Ward type (84%)	Admission triage & bed planning
Readmission Risk (RF)	AUC=0.736	Comorbidity count (21%)	Discharge risk flagging
Patient Segments (K-Means)	k=4, PCA validated	Age + severity profile	Personalised care pathways

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